

Spiking neural network solvers for model predictive control of autoclave composite curing

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Revision 3 — final software-phase report. Revision 1 was written against `snn_opt` 0.4.0, in which the formal convergence flag never fired at any horizon. Section 3.4 of that revision diagnosed the cause as an absolute, scale-sensitive stopping test and predicted, on the record, that a scale-invariant replacement would raise the convergence rate at short and medium horizons but would **still not fire at the working horizon**. Revision 2 reported the outcome of acting on that diagnosis. Every number below is measured under `snn_opt` 0.6.0 unless explicitly labelled 0.4.0, and every figure is traceable to a file through `results/artifact-index.md`. **Revision 3 closes the problem-formulation question** (Section 3.9–Section 3.11). It establishes that the constrained output has *relative degree 5*, so the first five gradient-constraint rows were never constraints at all; that the residual output clipping — named in Revision 2 as the largest remaining objection to any equivalence claim — was caused by a projection watchdog **aborting** the solve, and is now **eliminated (0 % of applied moves, down from 12.9 % in the stiff window)**; and that two other plausible causes, the constraint set and the slack-penalty scaling, are refuted by measurement. Two Revision-2 expectations did not survive: lengthening or shortening the constraint set does not affect convergence at all, and the reported convergence *rate* falls when dead rows are removed purely because the certificate's threshold scales with the problem. Both reversals are reported in place rather than dropped. The headline conclusion is *same problem, feasible everywhere, every applied move now solver-produced, convergence established on a substantial minority of steps and specifically absent where the controller matters most*. It is still not an equivalence claim. Step-by-step validation evidence: `docs/PHASE4_VALIDATION_REPORT.md`.

Abstract

Model predictive control (MPC) requires solving a constrained quadratic program (QP) at every control interval, which places a hard computational floor on the achievable sampling rate and motivates interest in solvers that map onto parallel, event-driven hardware. Recent work has shown that the time-averaged firing rates of certain inhibition-dominated spiking neural networks (SNNs) yield the solutions of linear and quadratic programs, suggesting that an SNN could serve as a drop-in QP solver inside an MPC loop. Here we test that proposition on a demanding physical problem: the autoclave cure of a thick-sectioned composite laminate, whose exothermic Arrhenius kinetics make the linearised prediction model unstable precisely where control is most critical. We implement a non-linear finite-difference digital twin, a classical CVXPY/OSQP MPC baseline, and an SNN-QP controller built on a projected-gradient/LIF solver, and we subject the claim of solver equivalence to direct measurement. We report six findings. First, establishing that two controllers solve the *same* QP is substantially harder than it appears: we identify and correct a one-step prediction-window error and an asymmetric Jacobian clamp, either of which silently invalidates a head-to-head comparison, and we unify both controllers onto a single canonical QP construction verified bit-identical across a full trajectory. Second, the per-step QP is *infeasible* at stiff exotherm steps under the natural hard-constraint formulation — not approximately, but algebraically: five uniformity-constraint rows carry a zero normal vector, three of them against a negative right-hand side. Third, once the problem is made feasible by an exact ℓ_1 penalty on the

state rows alone, the SNN reaches the optimum to within 5×10^{-4} °C of a reference solver while its convergence flag never fires, because the library's projected-gradient stopping test is absolute rather than scale-invariant and cannot trigger on a problem whose gradient scale is $\sim 10^{10}$. Fourth — the test of that diagnosis — replacing the criterion with a scale-invariant KKT certificate raises formal convergence from 0 % to 51 % / 47 % / 23 % across nominal, disturbance and stiff-exotherm operation, makes every step feasible, and reduces the maximum constraint residual by five orders of magnitude, while leaving the closed-loop trajectory essentially unchanged (overshoot 13.24 $\rightarrow$ 13.23 °C): the applied control was already right, and what improved was its certification. At the long horizon the new certificate still fails by a factor of 113, exactly as predicted before it was run. Fifth, we show the zero rows are not an accident of particular states but the plant's **relative degree**: the manipulated air temperature reaches the surface node only after four diffusion steps and the centre node after six, so the through-thickness gradient cannot be influenced for five samples and its first five constraint rows are identically zero at every operating point and every horizon — a textbook consequence of plant delay, and one that no choice of horizon can repair, as we confirm by finding the unsatisfiable row set identical for $N = 5, 10, 15$ and 20 . Sixth, the residual output clipping that remained the strongest objection to any equivalence claim is **eliminated**: it was caused not by solver inaccuracy but by a projection watchdog *aborting* the solve after roughly 130 of 8000 permitted iterations on half the stiff window, and restoring the budget removes every abort and takes clipping from 12.9 % to **0 %** there, at a cost of roughly 30 $\times$ the stiff-window solve time — time the previous configuration had simply not been spending. We conclude that the two controllers solve the same QP, that their applied controls agree to 0.707 °C RMS, that every applied move is now solver-produced rather than filter-corrected, and that formal convergence nonetheless remains weakest — 16 % — in the stiff exotherm window the controller exists to handle, where every non-converged step now exhausts its iteration allowance without meeting the certificate.

1 Introduction

Thick-sectioned polymer-matrix composites are cured in an autoclave under a prescribed temperature profile. The governing difficulty is that the resin's polymerisation is exothermic: past the gelation point the reaction liberates heat faster than the low-conductivity laminate can shed it, so the interior of the part can run away thermally even while its surface is being cooled. Standard practice uses open-loop ramp-and-hold recipes, which cannot react to this, and which consequently trade cycle time and part quality for safety margin.

Model predictive control is the natural remedy, since it can anticipate the exotherm over a prediction horizon and brake before the interior overshoots. Its cost is computational: MPC requires a constrained QP solution every sampling interval. This has motivated a broad search for QP solvers suited to specialised hardware, and in particular for solvers whose structure maps onto parallel or event-driven substrates rather than sequential CPUs.

A line of theoretical work has established that spiking neural networks are intimately connected to convex optimisation: the connectivity, thresholds, and time constants of a network of leaky integrate-and-fire (LIF) neurons can be placed in correspondence with the parameters of a linear or quadratic program, such that the network's output solves that program [2]. If this correspondence holds in practice, an SNN is not merely a biologically-inspired analogue of a solver — it *is* a solver, one that is natively event-driven and therefore a candidate for low-power neuromorphic or FPGA deployment.

This report asks whether that candidacy survives contact with a hard control problem. The question is not whether an SNN can solve *some* QP; it is whether an SNN can be substituted for a classical solver inside

a closed control loop, on a plant stiff enough to be interesting, and produce demonstrably the same control. We take the phrase "demonstrably the same" seriously, and much of what follows concerns the surprising difficulty of establishing it.

Our contributions are:

1. A complete simulation environment — non-linear plant twin, classical MPC baseline, and SNN-QP controller — in which both controllers are driven from a single shared QP construction, so that solver equivalence is testable rather than assumed (Section 2, Section 3.1).
2. The identification of two distinct failure modes that each invalidate a naive comparison and which are individually invisible in aggregate closed-loop metrics: a one-step prediction-window offset, and an asymmetric regularisation of the Arrhenius Jacobian (Section 3.2).
3. An algebraic proof that the per-step QP is genuinely infeasible through the exotherm under hard state constraints — later corroborated independently by the solver library itself, which names the same constraint row — reframing the solver's behaviour and establishing that no amount of preconditioning could have repaired it (Section 3.3).
4. A precise diagnosis of the residual non-convergence as an artefact of an absolute, rather than scale-invariant, stopping criterion, together with evidence that the returned solution is nonetheless correct (Section 3.4).
5. A pre-registered test of that diagnosis: the predicted consequences of adopting a scale-invariant certificate were written down, with the number they were derived from, *before* the change was made; both the predicted improvement and the predicted residual failure at the working horizon are confirmed (Section 3.5). We also report two upgrade hazards that left every aggregate metric looking plausible while the part failed to cure (Section 3.6) and the compute cost honestly — the SNN is $\sim 8.9\times$ slower than the reference solver on CPU (Section 3.7).

2 Methods

2.1 Plant model

The laminate and its tooling are discretised as a one-dimensional conduction problem with three composite nodes and four tooling nodes. The state is

$$\mathbf{x} = [T_{c,1}, T_{c,2}, T_{c,3}, T_{t,1..4}, \alpha_1, \alpha_2, \alpha_3]^\top \in \mathbb{R}^{10},$$

where T_c are composite temperatures, T_t tooling temperatures, and α the local degree of cure. The single actuator is the autoclave air temperature T_a , which enters at the outer tooling node.

Cure advances by the single-term autocatalytic Arrhenius law

$$\frac{d\alpha}{dt} = A_c \exp\left(-\frac{E_a}{RT}\right) \alpha^m (1 - \alpha)^n, \quad (1)$$

and liberates heat into the composite energy balance in proportion to da/dt . The plant is integrated by explicit finite differences at $T_E = 60$ s and is never linearised: it is the ground truth against which both controllers are judged. All physical parameters are taken from the reference formulation [1] and were verified term-by-term against its appendix.

Open-loop simulation reproduces the phenomenon of interest: under a static ramp-and-hold profile the part centre exceeds 140 °C, well above the surface temperature and into thermal degradation.

2.2 Model predictive control formulation

At each interval both controllers linearise the plant about the current operating point $(T, \bar{\alpha})$, obtaining a discrete-time model (A_p, B_p) . Differentiating (1) produces the Jacobian terms

$$\frac{\partial \dot{\alpha}}{\partial T} = f_0 \frac{E_a}{RT^2}, \quad \frac{\partial \dot{\alpha}}{\partial \alpha} = f_0 \left(\frac{m}{\alpha} - \frac{n}{1-\alpha} \right), \quad (2)$$

with f_0 the reaction rate at the operating point. The second expression has genuine poles at $\alpha \rightarrow 0$ and $\alpha \rightarrow 1$; we therefore evaluate it at α clamped to $[10^{-3}, 0.999]$. This regularises the *point at which the derivative is evaluated*, not the model, and is applied identically to both controllers.

The MPC problem over a horizon N penalises tracking error, control effort, and control slew, subject to actuator limits and a spatial-uniformity constraint:

$$\min_u \sum_{k=0}^{N-1} \|x_k - x_{\text{ref}}\|_Q^2 + Ru_k^2 + S(u_k - u_{k-1})^2 \quad (3)$$

$$\text{s. t. } x_{k+1} = A_p x_k + B_p u_k, \quad T_{a, \min} \leq u_k \leq T_{a, \max}, \quad |u_k - u_{k-1}| \leq \Delta_{\max}, \quad |x_k^{(1)} - x_k^{(3)}| \leq G_{\max}.$$

The final constraint bounds the centre-to-surface temperature difference and is what enforces a uniform, inside-out cure.

2.3 Canonical condensed form

Eliminating the state recursion yields a dense QP in the input sequence $z = [u_0, \dots, u_{N-1}]^T$:

$$\min_z \frac{1}{2} z^T H z + f^T z \quad \text{s. t. } A_{\text{ineq}} z \leq b_{\text{ineq}}, \quad (4)$$

with $H = 2(\Gamma^T Q \Gamma + R + D^T S D)$ and $f = 2\Gamma^T Q(\Phi x_0 - x_{\text{ref}}) - 2D^T S d_0$, where Φ and Γ are the free- and forced-response matrices and D is the first-difference operator. Row i of Φ is A_p^i , predicting x_i ; this indexing is load-bearing and is revisited in Section 3.2.

Both controllers call one implementation of (4). The CVXPY/OSQP baseline solves it directly. The SNN controller applies a Jacobi change of variables before solving, described next.

2.4 SNN-QP solver and preconditioning

The SNN solver minimises $1/2 x^T A x + b^T x$ subject to $Cx + d \leq 0$ by alternating a gradient step with a greedy projection onto violated constraint boundaries — the LIF analogue being that a neuron's membrane potential is integrated until it crosses threshold, at which point a spike applies a discrete corrective displacement [2]. Constraints map to thresholds; the constraint Gram matrix CC^T plays the role of the recurrent weight matrix. We use the `snn_opt` implementation [5] with its compiled backend, and we classify what it actually runs as an *approximate iterative solver* rather than a realisation of the continuous-time theory: the theory motivates the method but does not certify this discretisation on this problem, and the distinction is load-bearing for every claim below.

Because H in (4) is stiff, the SNN adapter applies Jacobi scaling with $D_s = \text{diag}(\sqrt{\text{diag}(H)})$:

$$\hat{H} = D_s^{-1} H D_s^{-1}, \quad \hat{g} = D_s^{-1} f, \quad z = D_s^{-1} \hat{z}, \quad (5)$$

with constraint rows subsequently normalised. This is an exact change of variables; the solution is mapped back by $z^* = \hat{z}^* / D_s$ before use, and all reported residuals and objectives are evaluated on the *original* problem (4), never on the scaled surrogate. We note that the reference solver applies its own internal equilibration, so (5) is not an asymmetry between the two controllers so much as an explicit statement of what the reference does implicitly.

2.5 Evaluation protocol

Head-to-head numbers are generated by a single harness driving two independent plant instances from identical initial conditions, with disturbances injected at the same point in the per-step sequence for both branches. This matters: two separately-written simulation scripts had injected the disturbance on opposite sides of the control computation, producing a one-step misalignment that quietly invalidated any step-wise comparison between them.

We distinguish four questions that are frequently conflated:

- **Feasibility** — does the returned point satisfy $A_{ineq} z \leq b_{ineq}$ to tolerance, measured on the original problem?
- **Optimality** — is its objective close to a reference solver's on the identical arrays?
- **Applied-move agreement** — does u_0 , the only quantity reaching the plant, match?
- **Formal convergence** — did the solver's own stopping criterion fire, and if not, why?

Only the third determines closed-loop behaviour; the others explain it. We report a lower objective attained at an infeasible point as a failure, never as a success.

3 Results

3.1 A first implementation, and two bugs that flattered it

An initial SNN-QP controller closed the loop and produced a superficially reasonable trajectory, but was worse than the baseline on every metric and roughly a hundred times slower (≈ 13 s per solve). Two defects were responsible.

The first was mundane: the solver was running its pure-Python reference path rather than its compiled kernel. Switching backends reduced a single solve from ≈ 5300 ms to ≈ 60 ms with bit-identical output.

The second was substantive. The condensed builder contained a stabilisation step that rescaled the prediction matrix whenever it became unstable,

```
rho = max(abs(eigvals(Ap))); if rho >= 1: Ap *= 0.98 / rho
```

The intent was conditioning; the effect was to delete the exotherm from the prediction. During gelation $\rho(A_p)$ genuinely exceeds 1 — the linearised model is *correctly* predicting thermal runaway, and that prediction is exactly the signal the controller needs in order to brake in time. With it suppressed, the controller held the autoclave at maximum temperature through the exotherm and overshoot by 20 °C against the baseline's 13 °C. Removing the rescaling restored correct braking.

The lesson generalises beyond this codebase: an unstable prediction model is not automatically a numerical problem to be suppressed. Here instability was the physics.

3.2 Establishing that two controllers solve the same problem

With those fixes the two controllers produced similar aggregate metrics, and it was tempting to declare equivalence. Direct measurement did not support it, for two independent reasons.

A one-step prediction window offset. The condensed builder constructed Φ with exponent $i+1$ rather than i , so its cost and constraints applied to states $x_1 \dots x_N$ while the baseline's applied to $x_0 \dots x_{N-1}$. Both windows yield well-posed QPs; both produce plausible trajectories. Reconstructing each window and solving both with the *same* reference solver, so that only the window varied, showed agreement with the live baseline to 0.018 °C for the correct window and disagreement of 1.20 °C for the shipped one — with the discrepancy concentrated, predictably, where the dynamics are least linear.

An asymmetric Jacobian clamp. The SNN controller bounded the exotherm Jacobian terms of (2) while the baseline did not. At a benign operating point this is negligible ($\max|\Delta A_p| \approx 10^{-4}$); at a gelation-like point it reaches ≈ 1851 . The two controllers were therefore predicting with materially different models exactly where the comparison mattered.

Both were removed by routing both controllers through one canonical construction (Section 2.3) and by making the clamp an explicit parameter defaulting to *off* on both. Verification: building both controllers' QPs from the same state at each of 160 trajectory steps, including the disturbance and the full exotherm, gives a maximum difference of exactly zero across H, f, A_{ineq}, b_{ineq} , the bounds, and the variable ordering.

We stress how weak aggregate metrics are as evidence here. Before the fixes, the two controllers matched to within a few degrees on peak overshoot while diverging by as much as 64 °C in applied control mid-trajectory. Summary statistics can agree while the underlying trajectories do not.

3.3 The stiff QP is infeasible, not merely ill-conditioned

With the comparison made fair, the SNN still exhausted its iteration budget at every step and returned points violating the constraints by large margins. The natural hypothesis, and the one we initially pursued, was conditioning: the condensed Hessian is assembled from weights spanning three orders of magnitude and compounded through I , whose entries involve A_p^k .

That hypothesis is false, and we can exclude it directly. Replacing the Jacobi scaling (5) with exact eigenwhitening, so that $\text{cond}(\text{hat } H) = 1.0$ — the best-conditioned Hessian obtainable — changed the final feasibility by less than 3 %. A perfectly conditioned Hessian did not help.

Sweeping 24 configurations across clamp setting, constraint form, horizon, and step size located the real cause. **All twelve hard-constrained configurations are reported infeasible by the reference solver**, at every horizon and under both prediction models. The mechanism is visible in the constraint offsets: the uniformity constraint at horizon step $N-1$ carries an offset $\propto G A_p^{N-1} x_0$, amplified by $\rho(A_p)^{N-1} \approx 4200$ at the gelation peak. The feasible set is empty. The cold-start point is violated by 2.5×10^5 before a single iteration executes.

The infeasibility is exact, not marginal. A solver certificate is evidence but not proof, so we established the result algebraically. Condensation writes the forced-response matrix I one block-row at a time, and block-row 0 is never written: x_0 is the pinned current state, so no future input can influence it, and the corresponding loop range is empty. The row is therefore the exact zero vector, structurally, for every state and at every step — not small, but identically zero. Uniformity row $k=0$ inherits it directly, and at the gelation operating point rows $k=1 \dots 4$ measure zero to machine precision as well. Rows $k=2,3,4$ pair a zero normal with a *negative* right-hand side, so those constraints read

$$0 \leq -2.25 \times 10^2, \quad (6)$$

which is false for every z in $\mathbb{R}^N$. This is a stronger statement than "the solver reports infeasible": it holds independently of any solver, any tolerance, and any scaling. It also settles the whitening result of the previous paragraph and disposes of the rescaling hypothesis entirely — rescaling multiplies the constraint normal, and zero times anything is zero.

The same mechanism resolves a separate anomaly. Through the stiff window the solver reported performing *no projections at all* while its scaled constraint violation stayed pinned near 2.25×10^{12} and the iterate grew by three orders of magnitude — a pattern consistent either with a genuinely idle projector or with a broken counter. It is the former, and it is a consequence of the same dead row: the library's projection step guards degenerate constraints with a $\|c_j\|^2 < 10^{-12}$ test and skips them *without* incrementing its counter and without altering the residual. The greedy selector therefore re-selects the same unfixable row on every one of 8000 iterations. One mechanism accounts for the zero counter, the frozen violation, and the diverging iterate simultaneously.

Independent corroboration. A later version of the solver library, developed with no knowledge of this analysis, refuses to construct the hard-form problem at all, raising "*constraint row 82 has a zero normal and $d > 0$: the problem is certifiably infeasible*". Row 82 is precisely the $k=2$ row derived above.

Introducing slack variables on the predicted-state rows only — actuator limits are real physical bounds and remain hard — with an exact ell_1 penalty [6] restores feasibility. The linear penalty term drives slacks to zero wherever the hard constraint is attainable (observed magnitude 10^{-29} at a benign state), so nothing is relaxed that could have been satisfied. A solver-independent slack linear program confirms the soft form is feasible at every horizon tested, with minimum total slack $\leq 3.9 \times 10^{-7}$. Applied identically to both controllers, this preserves the parity established in Section 3.2. We note that softening is not a workaround chosen for convenience: against a zero constraint normal it is the *only* repair available, because adding a slack column is the only way to give the row a coefficient on something that can move.

Throughout the remainder of this report we name which constraint form is meant. The hard form is infeasible and the soft form is feasible; both statements are true, and conflating them reads as a contradiction.

3.4 Correct answers under a mis-specified stopping test

With a feasible problem the SNN returns feasible points, and the closed-loop numbers improve substantially (Table 3). The formal convergence flag, however, still never fires.

The blocker is identifiable. The projected-gradient norm at the returned point is $\approx 1.66 \times 10^{10}$ and is unchanged by a six-fold increase in iteration budget. The solver tests this norm as an *absolute* quantity against a tolerance of 5×10^{-2} . On a problem whose gradient scale is 10^{10} , satisfying that test demands twelve orders of magnitude of reduction; the criterion is not scale-invariant and cannot fire regardless of solution quality.

To determine whether the returned point is nonetheless correct, we evaluated at a horizon short enough that the A_p^{N-1} amplification is mild, and compared against a reference solve of the identical problem:

Quantity	Value
Applied move u_0 vs. reference optimum	agree to 5×10^{-4} °C
Relative objective gap	-5.4×10^{-8}

Constraint residual	5.4×10^{-4} (feasible)
Projected-gradient norm, <i>relative</i> to gradient scale	9.7×10^{-3} — inside the 5×10^{-2} tolerance
Reported converged flag	False

On a well-posed stiff QP the SNN reaches the optimum, and a scale-invariant form of the solver's own criterion would have fired. The persistent negative flag is a property of the stopping test, not evidence of a wrong answer. This is, we suggest, the most transferable finding here: absolute tolerances on gradient norms are unsafe for condensed MPC problems, whose objective scale varies by orders of magnitude with the operating point.

A prediction, recorded before it was tested. The result above is at a horizon of 5, where the A_p^{N-1} amplification is mild, and it would be an overclaim to generalise it. We therefore measured the same *relative* projected-gradient norm at the working horizon of 20, where it is **0.449 to 0.670** depending on the gradient step size — an order of magnitude outside the 5×10^{-2} tolerance. On that basis we recorded the prediction that a scale-invariant test, however correctly specified, would **still not fire at the working horizon**, and that the defensible claim would remain "the applied move is correct where we can check it against a reference solver", never "a better tolerance converges everywhere". Section 3.5 reports the test of that prediction.

We flag one methodological point that produced a spurious disagreement in our own work. The relative norm depends on the gradient step size and is meaningless without it; two measurements that differ (0.449 versus 0.670) can both be correct and describe different configurations. We now report the norm swept across step sizes so it cannot be quoted without one.

3.5 Acting on the diagnosis: a scale-invariant certificate

Section 3.4 attributes the persistent negative flag to the stopping test rather than to the solution. That is a falsifiable claim, and the way to test it is to replace the test and see which quantities move. A later release of the solver library supplies exactly the required substitute: a scale-invariant Karush–Kuhn–Tucker cone certificate, selectable in place of the legacy absolute projected-gradient test. We adopted it under a before-and-after regression fingerprint that records, for each horizon and step size, the applied move, the convergence flag, the projection counts, the residuals, and a hash of the full solution vector.

The diagnosis is confirmed, in both directions.

	legacy absolute test	scale-invariant KKT
Formal convergence — nominal	0.0 %	51.3 %
— disturbance	0.0 %	46.9 %
— stiff exotherm window	0.0 %	22.6 %
Steps feasible enough to score an objective gap	48.1 % / 58.8 % / 45.2 %	100 % / 100 % / 100 %
Max constraint residual	1.55 / 2.07 / 1.55	3.5e−5 / 1.9e−5 / 1.9e−5
Applied moves corrected by the safety filter	13.1 % / 15.0 % / 29.0 %	3.8 % / 1.3 % / 12.9 %
Closed-loop overshoot	13.24 °C	13.23 °C
Peak cure gradient $\Delta\alpha$	0.3418	0.3417

recommended configuration.

Median total ms/step, nominal	legacy configuration	scale-invariant KKT
Reference (CVXPY / OSQP)	5.76	5.47
SNN-QP	113.75	48.66
Ratio	19.8×	8.9×

Table 2. Wall-clock comparison, single process, horizon 10, soft state constraints.

Under the legacy criterion the solver never terminated early, exhausting its full 8000-iteration budget at a near-constant ~ 114 ms per step. The scale-invariant certificate lets roughly half of nominal steps stop early, approximately halving the median; inside the stiff window it falls further, to 27.2 ms ($5.0\times$ the reference), because there the projection watchdog terminates the hardest solves sooner. QP *construction* costs the two sides the same (0.73 versus 0.74 ms); the entire gap is in the solve.

Two caveats. First, timing is the only quantity in this report that is not bit-reproducible: repeating the identical run moves the absolute medians by a few percent with machine load, while the ratio holds. Read the ratio. Second, an earlier claim in this project that compute time was "comparable to" the reference was drawn from a superseded long-horizon, hard-constraint harness in which the reference solver is itself slow (116 ms/step), flattering the ratio to $\sim 1.6\times$. That claim did not hold at the configuration actually recommended and has been withdrawn. No claim of computational advantage is made anywhere in this report; the architectural case for the SNN rests on event-driven hardware, which this work does not evaluate.

3.8 Closed-loop results, and a degenerate configuration

Table 3 gives the controlled comparison at the settled configuration.

Table 3. Controlled head-to-head, horizon 10, soft state constraints, identical prediction model, scale-invariant certificate. Trajectory difference is the Euclidean norm over composite temperatures and cure states.

Metric	Nominal	Disturbance	Exotherm window
RMS applied-control difference	0.707 °C	0.793 °C	0.565 °C
RMS trajectory difference	0.251	0.286	0.418
Max applied-control difference	3.52 °C	3.95 °C	0.66 °C
SNN max constraint residual	3.5×10^{-5}	1.9×10^{-5}	1.9×10^{-5}
SNN formal convergence rate	51.3 %	46.9 %	22.6 %
Steps feasible enough to score objective gap	100 %	100 %	100 %
Mean objective gap on feasible steps	6.7×10^{-4}	2.7×10^{-3}	2.6×10^{-3}
Applied moves corrected by safety filter	3.8 %	1.3 %	12.9 %
Final degree of cure (cure gate)	0.9999	0.9998	—

Relative to the pre-correction configuration, RMS applied-control difference falls from 16.005 °C to

0.707 °C and the maximum constraint residual from 1.85×10^5 to 3.5×10^{-5} — ten orders of magnitude. Both controllers achieve full uniform cure with comparable overshoot (13.77 °C baseline, 13.23 °C SNN) and identical actuator-limit compliance.

The objective-gap row is not comparable to the corresponding row of the legacy configuration; see the like-for-like restriction in Section 3.5. The convergence row is the honest weak point of the study, and its shape matters more than its average: the stiff exotherm window, the single regime that motivates predictive control of this process at all, is where the certificate fires least often and where the safety filter intervenes most.

A cautionary configuration. At horizon 5 the comparison appears flawless: RMS applied-control difference 0.000 °C, formal convergence 98.8 %, zero clipping, zero constraint violations. It is also useless. A five-minute horizon cannot see past the thermal transport lag, so the effort term dominates and the controller drives the actuator to its lower bound. In the nominal scenario the part cools from 28 °C to 11.4 °C; in the disturbance scenario, to 1.85 °C. The degree of cure reaches at most 1.0×10^{-5} — below reporting precision, though we note that it is not identically zero, and an earlier draft of this report said "never departs from zero", which the saved trace does not support. Both controllers agree perfectly because both are inert, saturated against the same bound. We report this because it is the precise failure mode that an equivalence study is most likely to mistake for success, and because it is invisible in every metric except the cure state itself — the same rule Section 3.6 reaches from the direction of a dependency upgrade. The same caution applies more weakly during heat-up, where 41.7 % of the first 60 steps — and 33.1 % of all 160, and 77.4 % inside the stiff window — have both controllers pinned at the slew limit; agreement on those steps reflects a shared actuator bound and is excluded from our claims.

3.9 The zero rows are the plant's relative degree

Section 3.3 established that certain gradient-constraint rows have an identically zero normal vector, and attributed this to the pinned initial state. That explanation is correct for $k=0$ but incomplete, and the complete version is more useful because it is structural.

The manipulated variable T_a enters the discretised plant at one node only — the outer tooling node, index 6 — so B_p has a single non-zero entry. Heat then diffuses inward one node per sample. Propagating $A_p^p B_p$ and recording where it first becomes non-zero:

p	support of $A_p^p B_p$	reaches T_{c3} (index 2)	reaches T_{c1} (index 0)
0	$\setminus\{6\}$	—	—
1	$\setminus\{5,6\}$	—	—
2	$\setminus\{4,5,6\}$	—	—
3	$\setminus\{3,4,5,6\}$	—	—
4	$\setminus\{2,\dots,6\}$	yes	—
6	$\setminus\{0,\dots,6\}$	yes	yes

The constrained output is $c^T x = x_0 - x_2 = T_{c1} - T_{c3}$, and the condensed prediction gives $F_{i,j} = A_p^{i-1-j} B_p$. Row i of the gradient block is therefore the zero vector precisely when $c^T A_p^p B_p = 0$ for all $p \leq i-1$, which by the table above means $i \leq 4$. The constrained output has **relative degree 5**, and

`gradientrowsk = 0, 1, 2, 3, 4` are exactly zero at every operating point and every horizon.

This is not a numerical near-degeneracy and not a property of stiff states: it follows from the sparsity pattern of the diffusion stencil, which does not change. It is also a documented MPC failure mode rather than a novel one. The MathWorks MPC Toolbox documentation states the same rule for output-variable constraints inside a plant's delay: for a plant with five sampling periods of delay, an output constraint before the sixth prediction step is in general impossible to satisfy, and all output constraints should therefore be softened [7].

Two consequences follow, and they explain both anomalies reported in Section 3.3 as one mechanism. A row with a zero normal is not a constraint on the decision variable at all; it reduces to $0 \leq \text{GRADIENT}_{\max} \text{ mp } (\Phi \ x_0)_k$, a predicate on the *current* state. When that predicate is false the QP is unconditionally infeasible — at any horizon, for any solver. And because `snn_opt` skips rows with squared norm below 10^{-12} *without* incrementing its projection counter or altering the residual, the projection selector re-picks the same dead row indefinitely, which is exactly the `n_projections = 0` observation.

We therefore omit these rows from the QP and report them separately. Omitting them without reporting would replace a real physical limitation — a predicted excursion the actuator cannot pre-empt within the transport delay — with silence. The reported quantity `unactionable_predicted_violation_degC` reaches 391 °C on the nominal run, and it must be read for what it is: a frozen-Jacobian prediction from a model we measure below to over-predict this very quantity by two orders of magnitude. The *actual* non-linear plant peaks at 28.9 °C and breaches the 10 °C limit on 3 of 160 steps.

Horizon length is not the mechanism and cannot be the remedy. The set of unsatisfiable rows is identical for $N = 5, 10, 15, 20$, measured at every step of a 159-step trajectory. Removing the dead rows resolves exactly two infeasible steps at every horizon; the remainder are live rows far out in the horizon, discussed in Section 3.11. One further degeneracy deserves recording: with $r=5$, an $N=5$ horizon has **no live gradient rows at all**, so its constraint set is vacuous. The configuration Section 3.8 identifies as degenerate for failing to cure the part is independently degenerate for silently discarding the only output constraint in the problem.

3.10 The residual clipping was an aborted solve, not an inaccurate one

Revision 2 named output clipping — 3.8 % of applied moves overall and 12.9 % inside the stiff window — as the largest remaining objection to any equivalence claim, and attributed it to solver inaccuracy. That attribution was wrong, and the correction removes the objection.

`snn_opt` reports a `convergence_reason` that distinguishes three terminations which a bare `converged = False` conflates. Classifying the 31 stiff-window states at the Revision-2 projection budget of 2000:

termination	count	meaning
<code>projection_budget_exhausted</code>	15	the solve was aborted part-way
<code>max_iterations</code>	11	ran the full allowance, certificate unmet
<code>converged(...)</code>	5	certificate met

On roughly half the stiff window the solver was not converging slowly — it was stopping after approximately 130 of its 8000 permitted iterations and returning whatever iterate it had reached. That

iterate is not guaranteed admissible, and the output filter was rescuing it. The clipping was a symptom of the watchdog, not of the method.

Raising the budget removes every abort, and the effect saturates sharply:

<code>max_projection_iters</code>	stiff convergence	median solve (ms)	aborts
2000	16.1 %	9.8	15
5000	25.8 %	297.9	0
20 000	25.8 %	296.9	0
100 000	25.8 %	303.6	0
500 000	25.8 %	299.5	0

5000 is a threshold rather than a tuning parameter: two further order-of-magnitude increases change nothing. Raising the *iteration* cap instead has no effect at all (8000, 30 000 and 100 000 all give 25.8 %, at up to twelve times the runtime), which localises the residual non-convergence to the method rather than to any budget. The benign window is unaffected by the budget in either direction, at 60 %.

In closed loop the payoff is unambiguous:

Table 4. Effect of the projection budget, closed loop, $N=10$, soft, k_0 -scale 0.1.

Metric	Rev. 2 (budget 2000)	Rev. 3 (budget 5000)
Applied moves from the safety clip, nominal	3.8 %	0.0 %
Applied moves from the safety clip, stiff	12.9 %	0.0 %
<code>projection_budget_exhausted</code>	~half of stiff steps	0
Max constraint residual	1.92×10^{-5}	6.8×10^{-7}
Median solve time, stiff	27.2 ms	425.9 ms

Every applied move is now produced by the solver rather than corrected by the filter. The cost is substantial and we state it without softening: overall compute rises from 8.9× to 16.5× the OSQP baseline, and within the stiff window from 5.0× to 43.7×. This is not a regression. The earlier figure was fast *because the solver was giving up*, and the corrected number is the true price of attempting the solve. A practitioner who prefers the earlier operating point can restore it, and inherits 12.9 % clipping with it.

We separate the two Revision-3 changes by experiment rather than asserting their effects:

configuration	stiff convergence	stiff clipping	residual
budget 2000, dead rows kept	22.58 %	12.90 %	1.92×10^{-5}
budget 2000, dead rows dropped	16.13 %	12.90 %	2.37×10^{-5}
budget 5000, dead rows dropped	16.13 %	0.00 %	6.84×10^{-7}

Dead-row removal moves the convergence rate and nothing else; the budget moves the clipping and the

residual.

The apparent convergence regression is a moving threshold, not a worse answer. The KKT tolerance is $\text{texttt{kkt_tol}} \times \text{texttt{kkt_scale}}$, and removing rows shrinks `kkt_scale`, tightening the test while the residual is unchanged. At step 80 the scale falls from 341 to 302 and the tolerance from 3.41×10^{-2} to 3.02×10^{-2} ; the applied move differs by 1.4×10^{-14} °C. The solution is the same to machine precision and the bar moved. This yields a reporting rule we now observe: **convergence rates are not comparable across different constraint sets**, because the certificate threshold is a function of the problem.

A related caution: the certificate is a *conjunction*, `converged(kkt(...); obj_plateau(...))`. Disabling early stopping removes the plateau term and convergence falls to 0 %, which we initially and wrongly read as evidence that the KKT test was never firing.

3.11 Two candidate remedies refuted, and where the difficulty actually lies

Two further explanations for the stiff-window convergence rate were plausible enough to test, and both fail. We record them because the cost of re-deriving a refuted hypothesis is high.

The constraint set does not drive convergence. Sweeping the number of imposed gradient rows from ten down to one — dead-row removal together with every constraint horizon N_c from 6 to N , at both $N=10$ and $N=20$ — leaves stiff-window convergence flat at 16.1 % and 19.4 % respectively, with the applied move unchanged to about 10^{-7} °C. The reason is visible in the conditioning: `_condition` row-normalises the constraint matrix, so the row-norm spread is $\sqrt{2}$ in *every* configuration, and $\text{cond}(H) \approx 780$ in both stiff and benign states. There is no conditioning headroom to recover. We therefore do not adopt a constraint horizon, though the option is implemented and documented.

The ℓ_1 penalty is already exact. Kerrigan and Maciejowski [6] give the exactness condition $\rho > l\text{Vert } \lambda^* r\text{Vert}_\infty$, with λ^* the multipliers of the hard problem on the softened rows. Measuring both sides across 159 steps: $\rho = 10^3$ against a maximum $l\text{Vert } \lambda^* r\text{Vert}_\infty$ of 1.15×10^{-5} and a median of 1.51×10^{-7} . The condition holds on **100 %** of the 146 hard-feasible steps, with a margin of roughly 10^8 , and the realised discrepancy between the soft and hard solutions on the applied move is 4.1×10^{-6} °C. Penalty scaling was never a defect. The quadratic slack term is never exact at finite weight by construction, and the figure above bounds its contribution.

The step size is likewise already at its best value: sweeping k_0 -scale over $0.05, 0.1, 0.5, 0.9$ gives 22.6 %, **25.8 %**, 19.4 % and 16.1 %, monotone away from the configured 0.1 in both directions.

Where the difficulty actually lies is the prediction model. The linearisation freezes the Arrhenius Jacobian at the current operating point, but the real exotherm is self-limiting: as $a \rightarrow 1$ the rate term collapses and the reaction burns out. A frozen Jacobian cannot represent its own extinction. Measured from step 88, with $\rho(A_p) = 1.43$:

Table 5. Predicted versus actual through-thickness gradient (°C), from step 88.

prediction step h	linear free response	actual non-linear
0	−3.58	−3.58
2	16.79	0.30
4	115.27	28.87

6	341.30	11.66
10	1798.54	6.78

The prediction is qualitatively right for about two steps, badly wrong by four, and absurd by ten. This is what generates the enormous slack values — up to 5.2×10^6 at $N=20$ — and, combined with Section 3.9, it produces an uncomfortable structural result: the constrained output cannot be influenced before step 5, while the prediction stops being quantitatively trustworthy after about step 3. **The controllable window opens after the trustworthy window has closed.** No choice of constraint horizon can satisfy both conditions during gelation, which is the deeper reason Section 3.11's first refutation holds.

We emphasise what this is not. It is not a statement that the physical constraint is unattainable, and it does not license shrinking A_p ; Section 3.1 already records that suppressing the unstable prediction erases the exotherm and costs 9 °C of overshoot. The unstable model is right about the direction of the excursion and wrong about its magnitude far out. Repairing that properly requires re-linearising along the predicted trajectory — an LTV rather than LTI prediction — which changes the condensation itself and is beyond the scope of this revision.

4 Discussion

We set out to test whether a spiking network can be substituted for a classical QP solver inside an MPC loop. Our answer is qualified in a specific way: the two controllers demonstrably receive the same problem, their applied controls agree to 0.707 °C RMS, every step of the trajectory is feasible with a maximum constraint residual of 7.6×10^{-7} , **every applied move is now produced by the solver rather than corrected by a downstream filter** (clipping 0 % in all three scenarios, from 12.9 % in the stiff window), and formal convergence — previously unattainable at any horizon that cures the part — is certified on 50 % of nominal steps. But convergence is **16.1 %** in the stiff exotherm window, every non-converged step there exhausts its iteration allowance without meeting the certificate, and at the long horizon the certificate fails by a factor of 113. The price of removing the clipping is a rise from $8.9 \times$ to $16.5 \times$ the reference solver's compute, and to $43.7 \times$ within the stiff window. We therefore state the result as *same problem, feasible everywhere, every applied move solver-produced, convergence established on a substantial minority of steps and weakest exactly where the controller matters*, and continue to decline to call it equivalence. We use no comparative term — "equivalent", "on par", "comparable" — anywhere in this report without an accompanying numerical tolerance.

It is worth being precise about what improved and what did not, because the two are easily conflated. What improved is the solver's *certification* of its own output: the flag, the residual, the feasibility coverage, and the frequency with which a downstream filter has to intervene. What did not measurably change is the *control*: overshoot moved by 0.01 °C and the peak cure gradient by 0.0001. A reader is entitled to ask whether the work therefore changed anything. It did, in the sense that matters for deployment — before, a correct applied move could not be distinguished from an incorrect one by any signal the solver emitted; now it can, on about half of steps. An uncertified correct answer and an uncertified wrong answer look identical from outside, and closing that gap is the prerequisite for trusting the loop rather than merely observing that it happens to work.

Five observations seem worth carrying forward.

Equivalence is an experimental claim with a high evidentiary bar. Two of the defects we found — the prediction-window offset and the asymmetric Jacobian clamp — produced closed-loop trajectories that looked entirely reasonable and aggregate metrics that were close. Neither is detectable without constructing both QPs and comparing them numerically. We would suggest that any claim of solver substitutability be accompanied by a per-step array-level comparison, not a metrics table.

Infeasibility masquerades as non-convergence. For a substantial portion of the trajectory the problem we posed had no solution, and every diagnostic we ran on the solver was, in retrospect, measuring the consequences of that. The signature — iteration budget exhausted, large residuals, lower-than-optimal objective at an infeasible point — is indistinguishable from a struggling solver until a reference solver is asked the same question. Condensed MPC formulations with state constraints and unstable prediction models appear especially prone to this, since the constraint offsets inherit $\rho(A_p)^{N-1}$ amplification while the constraint gradients do not.

Stopping criteria must be scale-invariant. The solver studied here reached the optimum and reported failure, because its convergence test compared an absolute gradient norm against a fixed tolerance on a problem whose natural scale is 10^{10} . This is a small implementation detail with a large consequence, and it will recur in any application where the objective scale varies with the operating point — which is to say, in most non-linear MPC. Replacing it was the single highest-value change in this project. But scale-invariance is necessary, not sufficient: the corrected certificate still does not fire at the working horizon, so the underlying iteration, not only its stopping rule, remains a limitation.

A metric that looks plausible is not a metric that is correct. Four separate failure modes in this study — a degenerate short horizon, a deprecated argument silently selecting the old criterion, a repurposed iteration limit aborting the solve, and an objective gap that appeared to regress — each left a table of aggregate numbers that a reader would accept. Three of the four were caught only by checking a quantity outside the table (the terminal degree of cure, the resolved solver configuration, the set of steps a mean was taken over). We would suggest that studies of this kind carry an explicit, machine-checked gate on a physical outcome, distinct from the metrics being reported, and that per-step records be retained so that a suspicious aggregate can be re-partitioned after the fact rather than re-measured.

A constraint the actuator cannot reach is not a constraint. The single most consequential defect in this study was imposing an output constraint inside the plant's transport delay, where the constraint row is identically zero and the QP is unconditionally infeasible. It survived three revisions because it is invisible in every aggregate metric and because the soft reformulation absorbed it. The general lesson is to check the *relative degree* of any constrained output against the row index at which it is first imposed, before attributing an infeasibility or a convergence failure to the solver. We would also suggest that a removed constraint row be reported rather than deleted: the row we dropped still encodes a real physical prediction, and reporting it is what distinguishes a formulation fix from a quiet relaxation.

Limitations. The horizon reduction that made the problem tractable was selected empirically rather than from the plant's thermal time constants. Softening the uniformity constraint weakens its guarantee, though the hard form it replaces is provably infeasible and therefore unimplementable; we mitigate this by reporting slack magnitudes and by verifying that the solution is insensitive to the penalty weight across two orders of magnitude. Our strongest evidence that the returned solution is optimal comes from a horizon too short to control the plant; at the working horizon we establish that the applied move matches the reference optimum to 1.5×10^{-6} °C but the full-horizon vector does not, and we do not claim otherwise. Formal convergence remains absent on half of nominal steps and 84 % of stiff ones, and the

residual is no longer attributable to any budget: every non-converged stiff step exhausts its iteration allowance with the certificate unmet, and three order-of-magnitude increases in that allowance change nothing. Output clipping, which earlier revisions named as the largest outstanding objection to any equivalence claim, is now 0 %; the cost is that compute time rises to 16.5× the reference on CPU overall and 43.7× within the stiff window. The prediction model remains a frozen Jacobian, which we measure over-predicting the constrained output by two orders of magnitude ten steps ahead during gelation; this, rather than the constraint or the penalty, is what generates the large slacks, and repairing it requires an LTV prediction we did not attempt. The formulation also has no terminal cost, terminal set or local controller, so we claim step-wise feasibility as observed and make no recursive-feasibility or nominal-stability claim. Finally, all results are in simulation against a digital twin of a single laminate geometry; no physical autoclave was involved.

Future work. Three of the four directions proposed in Revision 2 have now been tested and three are closed. Terminal or constraint-set redesign does not help: the constraint set does not influence convergence at all (Section 3.11), and the formulation has no terminal ingredient for a redesign to act on. Step-size re-tuning does not help: k_0 -scale 0.1 is already optimal of the values swept, so the conjecture that the previous value was tuned against the superseded criterion and had become stale is not supported. The saturating-projection question is settled: the watchdog made the pathology visible and did not repair it, and repairing it is what eliminated the clipping (Section 3.10).

What remains is narrower and harder. The binding limitation is the prediction model — a frozen Jacobian that over-states the constrained output by two orders of magnitude ten steps ahead during gelation (Table 5). Because the constrained output cannot be influenced before step 5 while the prediction ceases to be trustworthy after about step 3, no constraint horizon exists that is simultaneously actionable and accurate during the exotherm. The principled remedy is an LTV prediction, re-linearising along the predicted trajectory, which changes the condensation and is a substantially larger piece of work than anything attempted here. Constraint tightening in the manner of tube MPC would be the rigorous alternative. Adding terminal ingredients would, separately, supply the recursive-feasibility guarantee the controller currently lacks.

Hardware implementation should continue to wait, but the reason has narrowed. The clipping objection is gone: every applied move is now solver-produced. What remains is that the convergence certificate fires on 16 % of steps in the regime the controller exists for, and that this residual is a property of the iteration rather than of any budget we can raise. A reasonable gate would be stiff-window convergence comfortably above 90 %, on a configuration that still cures the part; clipping in the low single digits, the other half of the Revision-2 gate, is already met at 0 %. The architectural argument for event-driven hardware is unaffected by any of this and remains the motivation for the work — on CPU the SNN is strictly worse than a mature classical solver, and it is only on a substrate that exploits its parallelism that the comparison becomes interesting. That is exactly why the hardware phase should test the SNN's real value proposition rather than inherit a known weakness.

Broader impact

The nearer-term impact of work in this direction is on process efficiency: closed-loop cure control shortens cycle times and reduces scrap in composite manufacturing, with corresponding reductions in energy and material waste relative to conservative open-loop recipes. A longer-term impact concerns

where such controllers can run — solvers that map onto low-power event-driven hardware make predictive control viable on embedded and edge platforms that cannot support a conventional optimisation stack.

Two cautions attach. First, a controller that appears to track a reference correctly while its underlying solver has not converged is a genuine safety concern in any process where constraint satisfaction protects the product or the equipment, and the present work is largely an illustration of how easily that state can be mistaken for success. We would argue that convergence and feasibility should be logged and monitored in deployment, not merely checked in simulation. Second, the same argument that makes neuromorphic solvers attractive for industrial edge control makes them attractive for autonomous systems generally, including applications whose desirability is less clear; that trade-off is not specific to this work but is not absent from it either.

Reproducibility

Every quantitative claim in this report is backed by a file in the repository's `results/` directory, and `results/artifact-index.md` maps each reported figure to that file and to the script which regenerates it. No number here exists only in prose or in console output. The per-step records underlying every aggregate are retained as CSV, which is what made the like-for-like objective-gap check of Section 3.5 possible after the fact.

Runs record their own provenance: the git commit and working-tree state, the versions of the solver stack, whether the compiled backend was actually loaded, the resolved convergence criterion, the full shared configuration applied to both controllers, and the terminal degree of cure. Both controllers are constructed from a single configuration dictionary, so a configuration cannot be applied to one side and not the other. Dependency changes are gated by a before-and-after fingerprint capturing, per horizon and step size, the applied move, the convergence flag, projection counts, residuals, and a hash of the solution vector.

All results are deterministic given the configuration: there is no random number generator anywhere in the plant or either controller. The single exception is wall-clock timing, which varies with machine load; Section 3.7 reports the ratio for that reason.

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